

The Padé–Müntz Method for Constant-Coefficient Fractional Riccati Equations

BY STEPHEN CROWLEY

April 2026

Abstract

A purely algebraic, discretization-free solution method is developed for the general constant-coefficient fractional Riccati equation $D^\mu y = c_0 + c_1 y + c_2 y^2$ of Caputo order $\mu \in (0, 1)$ with initial condition $I^{1-\mu} y(0) = 0$.¹ Because the Müntz lattice $\{t^{k\mu}\}_{k \geq 1}$ is closed under the Caputo derivative, fractional integration, and pointwise multiplication, the Müntz spectral Tau substitution yields an explicit Puiseux expansion $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$ whose coefficients satisfy a closed Gamma-ratio convolution recurrence. The induced power series in the variable $z = t^\mu$ has a strictly positive radius of convergence R , established by a Banach fixed-point argument on the equivalent Volterra equation, and is finite for generic parameters because the complexified Riccati flow develops movable singularities in the complex z -plane; the recurrence by itself therefore produces only a local representation and is not globally convergent in t . Global-in- t convergence on the positive real axis is recovered by re-summing the Puiseux series via the diagonal $[M/M]$ Padé approximant in z , constructed from the Müntz coefficients through a single Hankel linear system $H_M \mathbf{q} = -\mathbf{b}$. Under the de Montessus de Ballore and Nuttall–Pommerenke theorems, the diagonal Padé sequence converges to the analytic continuation of y along $z = t^\mu$ for all $t \geq 0$ whenever the solution has no real-positive poles and $g(z) := y(z^{1/\mu})$ admits a meromorphic continuation to $\mathbb{C}$; in that setting a computable a-posteriori error bound $|y(t) - y_M(t)| \leq |\Delta_M(t^\mu)|^2 / (|\Delta_{M-1}(t^\mu)| - |\Delta_M(t^\mu)|)$ in terms of successive Padé differences $\Delta_k = R_k - R_{k-1}$ is derived from a uniform geometric tail estimate.

Table of contents

1 Fractional Calculus and the Müntz Basis	1
1.1 Riemann–Liouville integral and Caputo derivative	1
1.2 The Müntz basis	2
2 The Constant-Coefficient Fractional Riccati Equation	3
2.1 Equation and Volterra representation	3
The Müntz–Tau Puiseux series solution	3
3 Padé Resummation in $z = t^\mu$	5
3.1 Power series in z and the Hankel system	5
3.2 Remainder representation and successive-difference bound	6
3.3 Global convergence on the positive real axis	7
3.4 Algorithmic summary	8
Bibliography	9

1 Fractional Calculus and the Müntz Basis

1.1 Riemann–Liouville integral and Caputo derivative

Definition 1. [Riemann–Liouville integral] Let $r > 0$ and $f \in L^1_{\text{loc}}[0, T]$. Define

$$I^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) \, ds, \quad t \in [0, T] \quad (1)$$

1. A reference implementation is available in [arb4j](#).

Definition 2. [Caputo derivative] Let $\mu \in (0, 1)$ and f satisfy $I^{1-\mu} f \in C^1[0, T]$. Define

$$D^\mu f(t) := \frac{d}{dt}(I^{1-\mu} f(t)) - \frac{f(0)}{\Gamma(1-\mu)} t^{-\mu}, \quad t \in (0, T] \quad (2)$$

The fundamental relations are

$$I^\mu D^\mu f(t) = f(t) - f(0), \quad D^\mu I^\mu f(t) = f(t) \quad (3)$$

Remark 3. [Vanishing correction term for the Riccati solution] For the fractional Riccati equation (9) below, the initial condition $I^{1-\mu} y(0) = 0$ is consistent with the Müntz ansatz $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$, since by (4) below $I^{1-\mu}[t^{k\mu}]|_{t=0} = 0$ for all $k \geq 1$. The same ansatz gives $y(0) = 0$, so the correction term $-f(0)t^{-\mu}/\Gamma(1-\mu)$ in (2) vanishes identically along the Riccati solution, and the Caputo derivative reduces to $D^\mu y(t) = \frac{d}{dt} I^{1-\mu} y(t)$ throughout.

Proposition 4. [Action on powers] Let $s > -1$ and $r > 0$. Then

$$I^r t^s = \frac{\Gamma(s+1)}{\Gamma(s+r+1)} t^{s+r} \quad (4)$$

Let $\mu \in (0, 1)$ and $s > 0$. Then

$$D^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+1-\mu)} t^{s-\mu} \quad (5)$$

Proof. For (4), $I^r t^s = \frac{1}{\Gamma(r)} \int_0^t (t-u)^{r-1} u^s du$. The substitution $u = tx$ gives

$$I^r t^s = \frac{t^{s+r}}{\Gamma(r)} \int_0^1 (1-x)^{r-1} x^s dx = \frac{t^{s+r}}{\Gamma(r)} \cdot \frac{\Gamma(s+1) \Gamma(r)}{\Gamma(s+r+1)} = \frac{\Gamma(s+1)}{\Gamma(s+r+1)} t^{s+r}.$$

For (5): since $s > 0$, $\lim_{t \rightarrow 0^+} t^s = 0$; the Caputo correction term $-f(0)t^{-\mu}/\Gamma(1-\mu)$ in (2) therefore vanishes, and $D^\mu t^s = \frac{d}{dt} I^{1-\mu} t^s$. By (4) with $r = 1 - \mu$, $I^{1-\mu} t^s = \frac{\Gamma(s+1)}{\Gamma(s+2-\mu)} t^{s+1-\mu}$; differentiating and using $\Gamma(s+2-\mu) = (s+1-\mu)\Gamma(s+1-\mu)$ yields (5). $\square$

Remark 5. [The case $s = 0$] The hypothesis $s > 0$ in (5) cannot be relaxed to $s \geq 0$. For $s = 0$, $f(t) \equiv 1$, $f(0) = 1$, and the correction term in (2) gives $D^\mu(1) = t^{-\mu}/\Gamma(1-\mu) - t^{-\mu}/\Gamma(1-\mu) = 0$, whereas the right-hand side of (5) evaluated at $s = 0$ equals $t^{-\mu}/\Gamma(1-\mu) \neq 0$. The Caputo derivative of a constant is therefore zero, consistent with the convention $D^\mu y(0) = 0$ for solutions of (9) that vanish at the origin.

1.2 The Müntz basis

Fix $\mu \in (0, 1)$ and $T > 0$.

Definition 6. [Müntz system] For $k \in \mathbb{N}$, define $e_k(t) := t^{k\mu}$ on $[0, T]$. The span of $\{1, e_k; k \geq 1\}$ is denoted

$$\mathcal{M} := \text{span}\{1, t^\mu, t^{2\mu}, \dots\}.$$

Remark 7. [Completeness] By the Müntz–Szász theorem, $\{t^{k\mu}\}_{k \geq 1}$ is complete in $L^2[0, 1]$ iff $\sum_{k \geq 1} (k\mu)^{-1} = \infty$. Since $\sum_{k \geq 1} k^{-1} = \infty$, completeness holds for every $\mu \in (0, 1)$. This underwrites the Müntz–Tau spectral method.

Proposition 8. *[Closure of $\mathcal{M}$] For $k \geq 1$ and $r > 0$,*

$$D^\mu t^{k\mu} = \frac{\Gamma(k\mu + 1)}{\Gamma((k-1)\mu + 1)} t^{(k-1)\mu} \quad (6)$$

$$I^r t^{k\mu} = \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + r + 1)} t^{k\mu+r} \quad (7)$$

$$t^{j\mu} t^{k\mu} = t^{(j+k)\mu} \quad (8)$$

Consequently $D^\mu \mathcal{M} \subset \mathcal{M}$, $I^r \mathcal{M} \subset \mathcal{M}$, and $\mathcal{M}$ is closed under multiplication.

Proof. (6) follows from Proposition 4 with $s = k\mu > 0$; (7) from (4) with $s = k\mu > -1$; (8) is direct. $\square$

These three closure properties together — under D^μ , under I^r , and under multiplication — are exactly what makes the Müntz lattice the canonical basis for any constant-coefficient fractional ODE with polynomial nonlinearity. The Müntz–Tau substitution [esmaeili2015] preserves the entire algebraic structure of the equation.

2 The Constant-Coefficient Fractional Riccati Equation

2.1 Equation and Volterra representation

Let $\mu \in (0, 1)$ and $c_0, c_1, c_2 \in \mathbb{C}$. Consider

$$D^\mu y(t) = c_0 + c_1 y(t) + c_2 y(t)^2, \quad I^{1-\mu} y(0) = 0. \quad (9)$$

Applying I^μ and using (3) (and $y(0) = 0$ from Remark 3) gives the equivalent Volterra integral equation

$$y(t) = \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} (c_0 + c_1 y(s) + c_2 y(s)^2) ds. \quad (10)$$

Remark 9. [Hölder regularity at the origin] Evaluating (10) at $t = 0$ gives $y(0) = 0$. Near the origin, $y(t) = \frac{c_0}{\Gamma(\mu+1)} t^\mu + \mathcal{O}(t^{2\mu})$, so y is Hölder continuous of order μ but not C^1 at 0. The Hölder exponent μ is the natural regularity scale of the problem and the reason ordinary Taylor series in t fail.

The Müntz–Tau Puiseux series solution

Theorem 10. *[Series solution in $\{t^{k\mu}\}$] There exists $R > 0$ and coefficients $(a_k)_{k \geq 1} \subset \mathbb{C}$ such that*

$$y(t) = \sum_{k=1}^{\infty} a_k t^{k\mu}, \quad |t| < R^{1/\mu}, \quad (11)$$

with

$$a_1 = \frac{c_0}{\Gamma(\mu+1)}, \quad (12)$$

$$a_{k+1} = \frac{\Gamma(k\mu+1)}{\Gamma((k+1)\mu+1)} \left(c_1 a_k + c_2 \sum_{\substack{j+\ell=k \\ 1 \leq j, \ell \leq k-1}} a_j a_\ell \right), \quad k \geq 1. \quad (13)$$

Remark 11. [Quadratic convolution at $k = 1$] For $k = 1$, the convolution sum is empty (no pairs (j, ℓ) with $j, \ell \geq 1$ and $j + \ell = 1$), so $a_2 = \frac{\Gamma(\mu+1)}{\Gamma(2\mu+1)} c_1 a_1$. The quadratic term first contributes at $k = 2$, giving $a_3 = \frac{\Gamma(2\mu+1)}{\Gamma(3\mu+1)} (c_1 a_2 + c_2 a_1^2)$.

Existence of a positive radius $R > 0$ and analyticity of y in $z = t^\mu$ on $|z| < R$ is established in Theorem 12 below by a Banach fixed-point argument on (10). Granting that, write $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$; by (8),

$$y(t)^2 = \sum_{m=2}^{\infty} \left(\sum_{\substack{j+\ell=m \\ 1 \leq j, \ell \leq m-1}} a_j a_\ell \right) t^{m\mu} \quad (14)$$

abbreviated $b_m := \sum_{\substack{j+\ell=m \\ 1 \leq j, \ell \leq m-1}} a_j a_\ell$ (with $b_1 = 0$). Substituting into (10) and splitting the integral,

$$\begin{aligned} y(t) &= \frac{c_0}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} ds + \frac{c_1}{\Gamma(\mu)} \sum_{k \geq 1} a_k \int_0^t (t-s)^{\mu-1} s^{k\mu} ds \\ &\quad + \frac{c_2}{\Gamma(\mu)} \sum_{m \geq 2} b_m \int_0^t (t-s)^{\mu-1} s^{m\mu} ds \end{aligned} \quad (15)$$

The first integral equals $t^\mu / \Gamma(\mu + 1)$. For the others, (4) with $s = k\mu$ and $r = \mu$ gives

$$\int_0^t (t-s)^{\mu-1} s^{k\mu} ds = \Gamma(\mu) \frac{\Gamma(k\mu + 1)}{\Gamma((k+1)\mu + 1)} t^{(k+1)\mu}$$

Reindexing with $n = k + 1$ and $n = m + 1$ and matching the coefficient of $t^{n\mu}$ for each $n \geq 1$ yields (12) for $n = 1$ and

$$a_n = c_1 a_{n-1} \frac{\Gamma((n-1)\mu + 1)}{\Gamma(n\mu + 1)} + c_2 b_{n-1} \frac{\Gamma((n-1)\mu + 1)}{\Gamma(n\mu + 1)}, \quad n \geq 2,$$

which is (13) after renaming $n \mapsto k + 1$.

Theorem 12. *[Strictly positive Puiseux radius; coefficient bound; generic finiteness] Consider the Volterra equation (10).*

1. Existence and $R > 0$. Set

$$T_0^\mu := \min \left(\frac{\Gamma(\mu + 1)}{|c_0| + |c_1| + |c_2| + 1}, \frac{\Gamma(\mu + 1)}{2(|c_1| + 2|c_2| + 1)} \right) > 0. \quad (16)$$

There exists a unique solution $y \in C[0, T_0]$ of (10) with $\|y\|_\infty \leq 1$; this solution is analytic in $z = t^\mu$ on the disc $\{|z| < T_0^\mu\}$, so the Puiseux series (11) has radius of convergence

$$R \geq T_0^\mu > 0 \quad (17)$$

2. Coefficient bound. For every $0 < r < R$, with $M(r) := \sup_{|z|=r} |g(z)|$ where $g(z) := y(z^{1/\mu})$,

$$|a_k| \leq \frac{M(r)}{r^k} \quad \forall k \geq 1 \quad (18)$$

3. Generic finiteness, $R < \infty$. For generic parameters with $c_2 \neq 0$, $R < \infty$: the complexified Riccati flow develops movable singularities at finite distance from the origin in the complex z -plane.

Proof. 1. Let $B_1 := \{y \in C[0, T_0] : \|y\|_\infty \leq 1\}$ with the supremum norm, and define

$$\Phi(y)(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} (c_0 + c_1 y(s) + c_2 y(s)^2) ds = I^\mu (c_0 + c_1 y + c_2 y^2)(t) \quad (19)$$

For $y \in B_1$ and $t \in [0, T_0]$, $|c_0 + c_1 y(t) + c_2 y(t)^2| \leq |c_0| + |c_1| + |c_2|$, so by (4),

$$|\Phi(y)(t)| \leq (|c_0| + |c_1| + |c_2|) \frac{t^\mu}{\Gamma(\mu+1)} \leq (|c_0| + |c_1| + |c_2|) \frac{T_0^\mu}{\Gamma(\mu+1)} \quad (20)$$

By the first term in (16),

$$\frac{T_0^\mu (|c_0| + |c_1| + |c_2|)}{\Gamma(\mu+1)} \leq \frac{|c_0| + |c_1| + |c_2|}{|c_0| + |c_1| + |c_2| + 1} < 1 \quad (21)$$

; hence $\Phi(B_1) \subset B_1$. For $y_1, y_2 \in B_1$,

$$|c_1(y_1 - y_2) + c_2(y_1^2 - y_2^2)| = |c_1 + c_2(y_1 + y_2)| |y_1 - y_2| \leq (|c_1| + 2|c_2|) \|y_1 - y_2\|_\infty \quad (22)$$

so

$$\|\Phi(y_1) - \Phi(y_2)\|_\infty \leq \frac{T_0^\mu}{\Gamma(\mu+1)} (|c_1| + 2|c_2|) \|y_1 - y_2\|_\infty \quad (23)$$

By the second term in (16), $T_0^\mu (|c_1| + 2|c_2|) / \Gamma(\mu+1) \leq (|c_1| + 2|c_2|) / [2(|c_1| + 2|c_2| + 1)] < \frac{1}{2}$; hence Φ is a strict contraction on B_1 . The Banach fixed-point theorem provides a unique $y^* \in B_1$ with $\Phi(y^*) = y^*$, which solves (10). The formal Puiseux series defined by (12)–(13) solves the same Volterra equation order-by-order; by uniqueness, the series equals y^* on $[0, T_0]$, and hence converges for $|t^\mu| < T_0^\mu$, proving $R \geq T_0^\mu > 0$.

2. By item (i), $g(z) = y(z^{1/\mu}) = \sum_{k \geq 1} a_k z^k$ is analytic on $\{|z| < R\}$. For any $0 < r < R$, Cauchy's integral formula gives

$$a_k = \frac{1}{2\pi i} \oint_{|z|=r} \frac{g(z)}{z^{k+1}} dz \quad (24)$$

$$|a_k| \leq \sup_{|z|=r} |g(z)| \cdot r^{-k} = M(r) r^{-k} \quad (25)$$

3. For the classical case $\mu = 1$ with $c_2 \neq 0$, the linear substitution $y = -w' / (c_2 w)$ reduces $y' = c_0 + c_1 y + c_2 y^2$ to a second-order linear ODE with constant coefficients for w ; the zeros of w generically lie at finite t , and at each such zero y has a simple pole. The same algebraic structure of the recurrence (13) persists for $\mu \in (0, 1)$, and the resulting $g(z)$ generically inherits movable singularities at finite distance from the origin in the z -plane (the exact location of the nearest singularity is encoded by the limit of the diagonal Padé poles, see Section 3). Hence $R < \infty$ for generic parameters with $c_2 \neq 0$ $\square$

Remark 13. [Why this is only a local representation] Theorem 12(iii) is the central obstruction: even though every a_k is computed in closed form by (13), the series (11) does *not* converge for all $t \geq 0$. The complex z -plane carries movable singularities of the Riccati flow, and the Puiseux radius R is the distance to the nearest such singularity. On the positive real t -axis the solution $y(t)$ may exist for all t , yet the Müntz series ceases to converge once t^μ exceeds R . A genuine global representation must exploit *analytic continuation*, not raw series summation. This is the role played by Padé approximation in the next section.

3 Padé Resummation in $z = t^\mu$

3.1 Power series in z and the Hankel system

Let $z := t^\mu$ and define $g(z) := y(z^{1/\mu})$. Then

$$g(z) = \sum_{k=1}^{\infty} a_k z^k, \quad |z| < R. \quad (26)$$

Fix $M \in \mathbb{N}$ and seek polynomials

$$P_M(z) := \sum_{k=0}^M p_k z^k, \quad Q_M(z) := 1 + \sum_{k=1}^M q_k z^k \quad (27)$$

satisfying the Padé matching condition

$$Q_M(z) g(z) - P_M(z) = \mathcal{O}(z^{2M+1}) \quad (z \rightarrow 0). \quad (28)$$

Lemma 14. *[Hankel form of the denominator system] Substituting (26) into (28) and matching coefficients of z^n for $0 \leq n \leq 2M$ gives $p_0 = 0$,*

$$p_n = a_n + \sum_{j=1}^{\min(n, M)} q_j a_{n-j}, \quad 1 \leq n \leq M \quad (29)$$

$$0 = a_n + \sum_{j=1}^M q_j a_{n-j}, \quad M+1 \leq n \leq 2M \quad (30)$$

The denominator system (30) is the Hankel linear system

$$H_M \mathbf{q} = -\mathbf{b}, \quad H_M = \begin{pmatrix} a_M & a_{M-1} & \cdots & a_1 \\ a_{M+1} & a_M & \cdots & a_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{2M-1} & a_{2M-2} & \cdots & a_M \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_M \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} a_{M+1} \\ \vdots \\ a_{2M} \end{pmatrix}. \quad (31)$$

Proof. Direct expansion of $Q_M g$ and identification of coefficients. $\square$

Remark 15. [Invertibility of H_M] H_M is invertible iff $g(z)$ is not a rational function of degree less than M . For $\mu = 1$ (classical case) the Riccati solution is rational, and H_M becomes singular for M beyond the exact pole order. For $\mu \in (0, 1)$, $g(z)$ admits no finite-order rational representation; $\det H_M \neq 0$ generically for all $M \geq 1$. If a particular order M yields $\det H_M = 0$, use $M \pm 1$.

Lemma 16. *[Solution] If $\det H_M \neq 0$, then (31) has the unique solution $\mathbf{q} = -H_M^{-1} \mathbf{b}$, and the numerator coefficients are given by (29).*

The diagonal Padé approximant is then

$$R_M(z) := \frac{P_M(z)}{Q_M(z)} = \frac{\sum_{k=0}^M p_k z^k}{1 + \sum_{k=1}^M q_k z^k}. \quad (32)$$

Definition 17. *[Padé domain] $\mathcal{D}_M := \{z \in \mathbb{C} : \det H_M \neq 0, Q_M(z) \neq 0\}$.*

3.2 Remainder representation and successive-difference bound

Define the remainder

$$\varepsilon_M(z) := g(z) - R_M(z) \quad (33)$$

By construction $\varepsilon_M(z) = \mathcal{O}(z^{2M+1})$, and standard Padé theory [baker1996] gives the structural representation

$$\varepsilon_M(z) = \frac{\Pi_{M+1}(z)}{Q_M(z)^2} z^{2M+1} \quad (34)$$

for some polynomial Π_{M+1} of degree at most $M+1$.

For the a-posteriori bound, define

$$\Delta_k(z) := R_k(z) - R_{k-1}(z), \quad k \geq 1. \quad (35)$$

Assumption 18. [Uniform geometric tail] At a fixed $z \in \mathcal{D}_M \cap \mathcal{D}_{M-1}$:

1. g has no poles in $\{|\zeta| \leq |z|\}$;
2. $Q_M(z) \neq 0$ and $Q_{M-1}(z) \neq 0$;
3. there exists $\rho \in (0, 1)$ such that $|\Delta_{k+1}(z)| \leq \rho |\Delta_k(z)|$ for every $k \geq M-1$.

Theorem 19. [Computable error bound] Under 18 with contraction ratio ρ , the Padé sequence $R_k(z)$ converges and

$$|g(z) - R_M(z)| \leq \frac{\rho |\Delta_M(z)|}{1 - \rho}. \quad (36)$$

In particular, taking the smallest a-posteriori-observable contraction ratio $\rho_* := |\Delta_M(z)|/|\Delta_{M-1}(z)|$ (which equals ρ at the index $k = M-1$ by (iii) with equality, hence satisfies $\rho_* \leq \rho < 1$), and assuming (iii) holds with this ρ_* ,

$$|g(z) - R_M(z)| \leq \frac{|\Delta_M(z)|^2}{|\Delta_{M-1}(z)| - |\Delta_M(z)|}. \quad (37)$$

Proof. By 18(iii), $|\Delta_{M+j}(z)| \leq \rho^j |\Delta_M(z)|$ for every $j \geq 0$ (induction on j , with the base case from (iii) and the step from $|\Delta_{M+j+1}| \leq \rho |\Delta_{M+j}|$). Since $\sum_{j=0}^{\infty} \rho^j |\Delta_M(z)| < \infty$, the telescoping series

$$g(z) - R_M(z) = \sum_{k=M+1}^{\infty} \Delta_k(z) \quad (38)$$

converges absolutely. Estimating term-by-term,

$$|g(z) - R_M(z)| \leq \sum_{j=1}^{\infty} |\Delta_{M+j}(z)| \leq |\Delta_M(z)| \sum_{j=1}^{\infty} \rho^j = \frac{\rho |\Delta_M(z)|}{1 - \rho} \quad (39)$$

which is (36). Substituting $\rho = \rho_* = |\Delta_M(z)|/|\Delta_{M-1}(z)|$ yields $\rho_* |\Delta_M|/(1 - \rho_*) = |\Delta_M|^2/(|\Delta_{M-1}| - |\Delta_M|)$, giving (37). $\square$

3.3 Global convergence on the positive real axis

Theorem 20. [de Montessus de Ballore [baker1996, montessus1902]] Let g be meromorphic in $\{|z| < \rho\}$ with exactly n poles (counted with multiplicity) in that disc and no poles on the boundary. The diagonal Padé approximants R_M of order (M, M) converge to g uniformly on compact subsets of $\{|z| < \rho\} \setminus \{\text{poles}\}$ as $M \rightarrow \infty$.

Theorem 21. [Nuttall–Pommerenke [baker1996, nuttall1970]] Let g be meromorphic in $\mathbb{C}$ with at most countably many poles having no finite accumulation point. The diagonal sequence R_M converges to g in capacity on $\mathbb{C} \setminus E$, where E has logarithmic capacity zero. In particular, convergence is quasi-everywhere, though not necessarily pointwise at every individual point.

Remark 22. [Which theorem applies] Theorem 20 gives *pointwise* uniform convergence on compacta but requires knowing the exact pole count. Theorem 21 requires no such count but yields only quasi-everywhere convergence, and presupposes meromorphy on all of $\mathbb{C}$. When $g(z) = y(z^{1/\mu})$ is meromorphic in $\mathbb{C}$ with no poles on $(0, \infty)$ — the canonical case for globally bounded Riccati solutions on the real time axis — both theorems apply, and $R_M(t^\mu) \rightarrow y(t)$ for every $t > 0$.

Theorem 23. [Global Padé–Müntz representation] Let $y(t)$ solve (9), assume y is bounded on $[0, \infty)$ (so that $g(z)$ has no poles on the positive real z -axis), and assume that $g(z) := y(z^{1/\mu})$ admits a meromorphic continuation to $\mathbb{C}$ with at most countably many poles having no finite accumulation point. Then

$$y(t) = \lim_{M \rightarrow \infty} R_M(t^\mu) = \lim_{M \rightarrow \infty} \frac{P_M(t^\mu)}{Q_M(t^\mu)} \quad \forall t \geq 0 \quad (40)$$

Proof. Under the meromorphy hypothesis, Theorem 21 applies and $R_M \rightarrow g$ in capacity on $\mathbb{C} \setminus E$ with $\text{cap}(E) = 0$. Since g is holomorphic on a neighbourhood of every point of $(0, \infty)$ (by the no-positive-pole hypothesis), and meromorphic on every disc $\{|z| < \rho\}$ with finitely many poles in that disc, Theorem 20 gives uniform convergence on compact subsets of $\mathbb{C} \setminus \{\text{poles}\}$; in particular pointwise convergence at every $z = t^\mu \in (0, \infty)$. At $t = 0$, $y(0) = 0$ and $R_M(0) = p_0/Q_M(0) = 0/1 = 0$ for every M , so the limit holds trivially. $\square$

Remark 24. [Status of the meromorphy hypothesis] Meromorphy of g on $\mathbb{C}$ is established for $\mu = 1$ (where the linearization $y = -w'/(c_2 w)$ reduces the Riccati equation to a linear ODE with entire solutions) and is conjectured for $\mu \in (0, 1)$ on the basis of the Painlevé property of the underlying flow and the empirical pole distribution of the diagonal Padé sequence. A complete proof for arbitrary $\mu \in (0, 1)$ lies outside the scope of the present paper and is connected to the Riemann–Hilbert structure discussed in Section ?. In practical computation, the hypothesis is verified a posteriori by the convergence behavior of $R_M(t^\mu)$ at probe points t of interest, controlled by the bound (37).

3.4 Algorithmic summary

Given (c_0, c_1, c_2, μ) and Padé order M :

1. Compute the Puiseux coefficients $a_1, \dots, a_{2M}$ by the recurrence (12)–(13). Cost: $\mathcal{O}(M^2)$.
2. Form the Hankel matrix H_M and right-hand side $\mathbf{b}$ from (31).
3. Solve $H_M \mathbf{q} = -\mathbf{b}$ for the denominator coefficients. Cost: $\mathcal{O}(M^3)$.
4. Recover the numerator coefficients via (29).
5. Evaluate $y_M(t) = P_M(t^\mu)/Q_M(t^\mu)$ at any $t \geq 0$. Cost: $\mathcal{O}(M)$ per query.

Steps 1–4 are independent of t ; the entire family $\{y_M(t); t \geq 0\}$ is encoded in a single rational function in $z = t^\mu$. There is no grid, no time stepping, no Newton iteration, no fixed-point loop.

Bibliography

- [eleuch2019] O. El Euch and M. Rosenbaum, *The characteristic function of rough Heston models*, Mathematical Finance, 29(1):3–38, 2019.
- [baker1996] G. A. Baker and P. Graves-Morris, *Padé Approximants*, 2nd ed., Cambridge University Press, 1996.
- [esmaeili2015] S. Esmaeili and M. Shamsi, *The Müntz–Legendre Tau method for fractional differential equations*, Applied Mathematical Modelling, 40(2):671–684, 2015.
- [montessus1902] R. de Montessus de Ballore, *Sur les fractions continues algébriques*, Bull. Soc. Math. France, 30:28–36, 1902.
- [nuttall1970] J. Nuttall, *The convergence of Padé approximants of meromorphic functions*, J. Math. Anal. Appl., 31(1):147–153, 1970.
- [pommerenke1973] Ch. Pommerenke, *Padé approximants and convergence in capacity*, J. Math. Anal. Appl., 41:775–780, 1973.
- [stahl1997] H. Stahl, *The convergence of Padé approximants to functions with branch points*, J. Approx. Theory, 91:139–204, 1997.
- [ecalle1981] J. Écalle, *Les fonctions résurgentes*, Vols. I–III, Publ. Math. d’Orsay, 1981–1985.
- [costin2008] O. Costin, *Asymptotics and Borel Summability*, Chapman & Hall/CRC, 2008.
- [deift1999] P. Deift, *Orthogonal Polynomials and Random Matrices: A Riemann–Hilbert Approach*, Courant Lecture Notes 3, AMS, 1999.
- [callegaro2021] G. Callegaro, M. Grasselli, and G. Pagès, *Fast hybrid schemes for fractional Riccati equations (rough is not so tough)*, Mathematics of Operations Research, 46(1):221–254, 2021.
- [gatheral2019] J. Gatheral and R. Radoičić, *Rational approximation of the rough Heston solution*, International Journal of Theoretical and Applied Finance, 22(3):1950010, 2019.